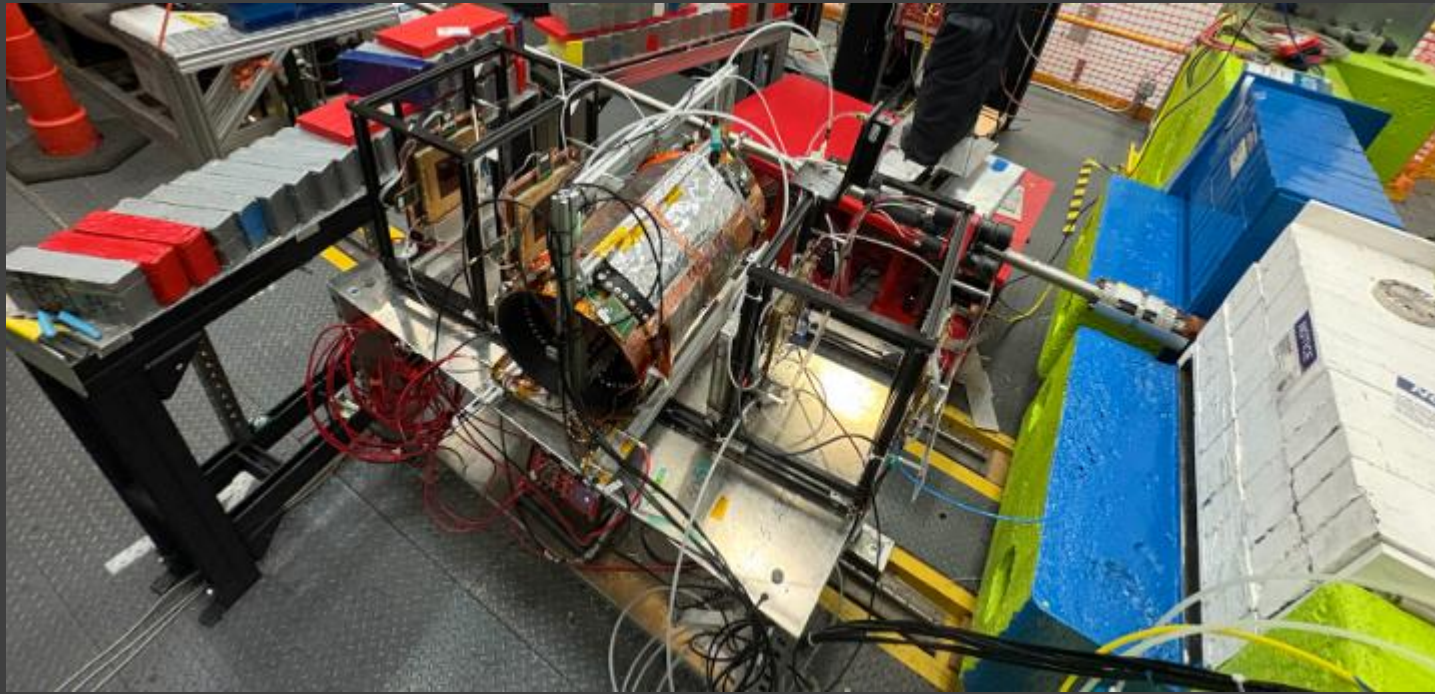




Mapping for a Cylindrical Micro-Resistive Well Detector

Semester 2 Project Plan

People

Group Members

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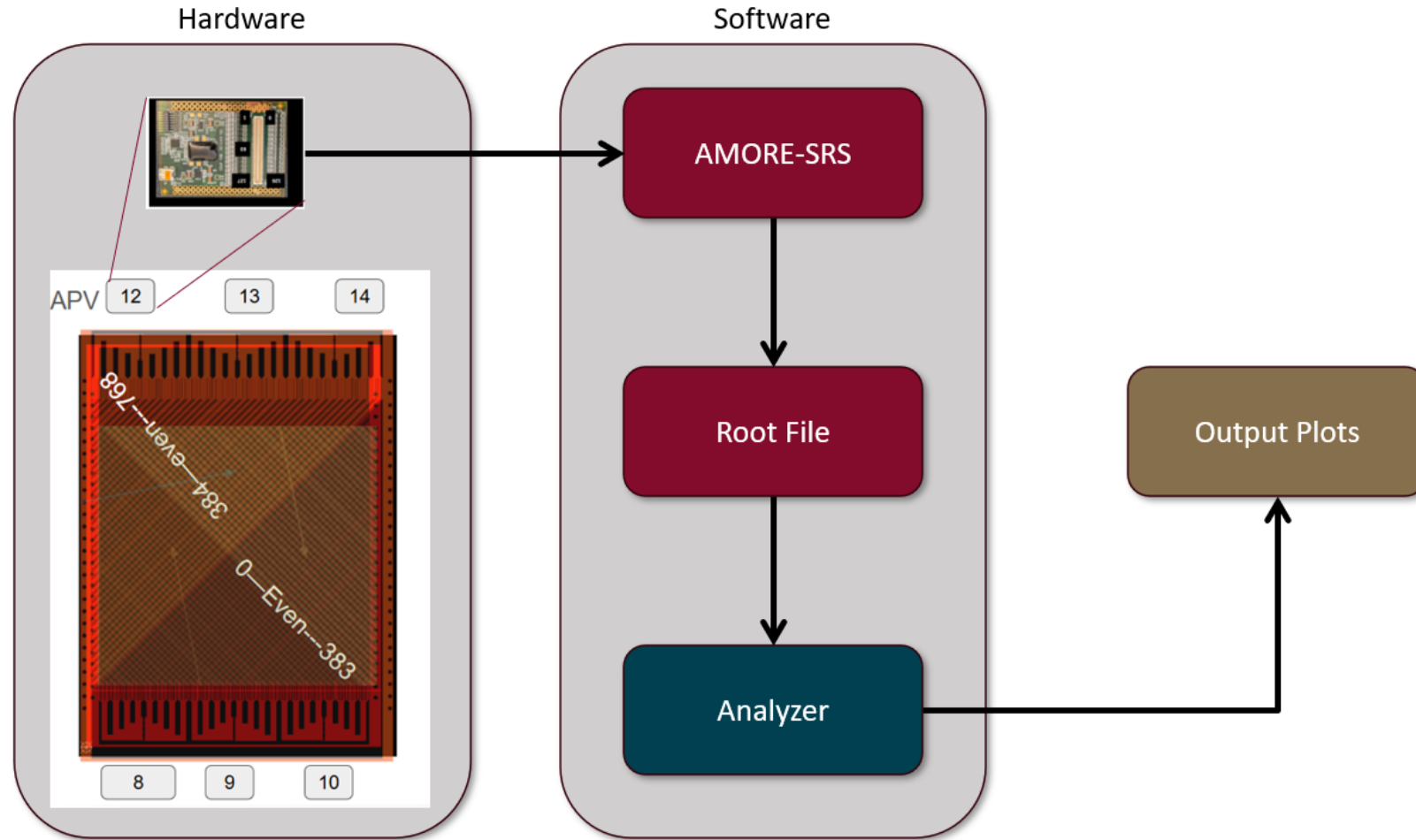
Faculty Advisors/Clients

- Pietro Iapozzuto,
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- Dr. Marcus Hohlmann,
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Goal and Motivation

The goal for the project is to create a program that characterises the efficiency and resolution for the cylindrical micro resistive well detector. The motivation for the project is to provide support for the software integration so the data can be more easily visualized and analyzed by the physicists.

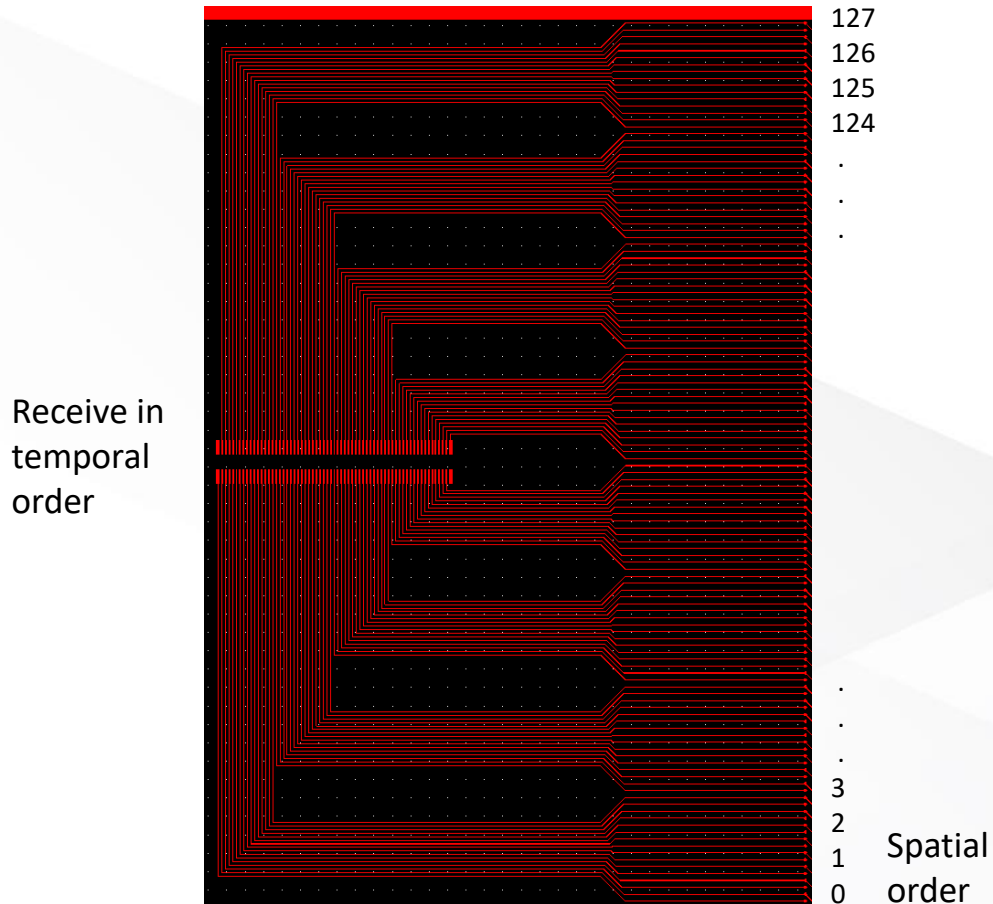
Approach



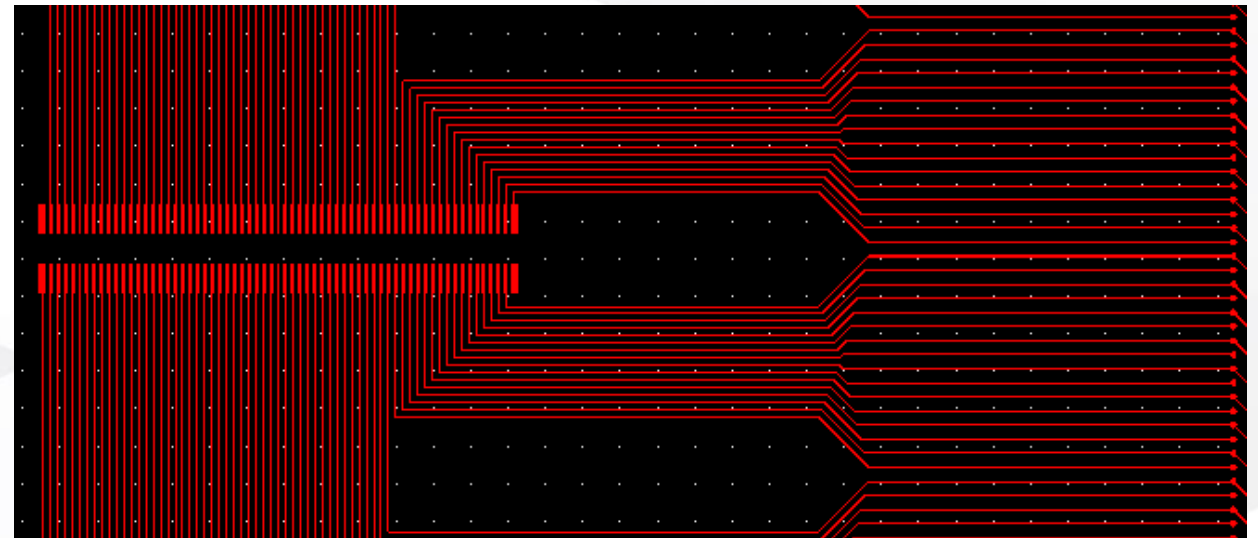
Approach

- Understand root software and existing progress
- Establish APV mapping functions and apply to cylindrical data
- Generate resolution plots
- Automate pipeline for multiple files and output plot types
- Verify correctness with efficiency plots

Approach- Generate mapping functions

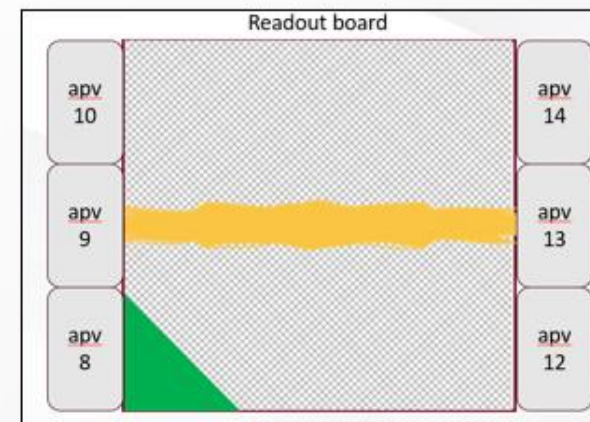
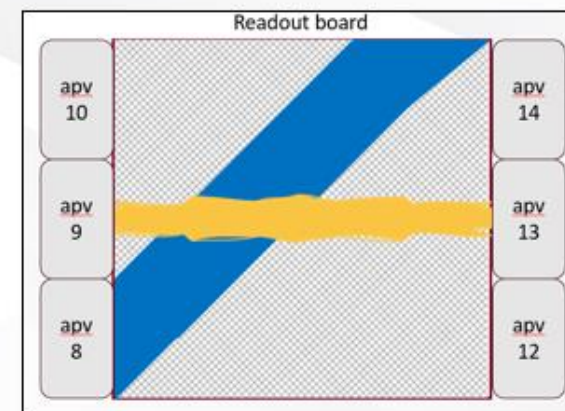
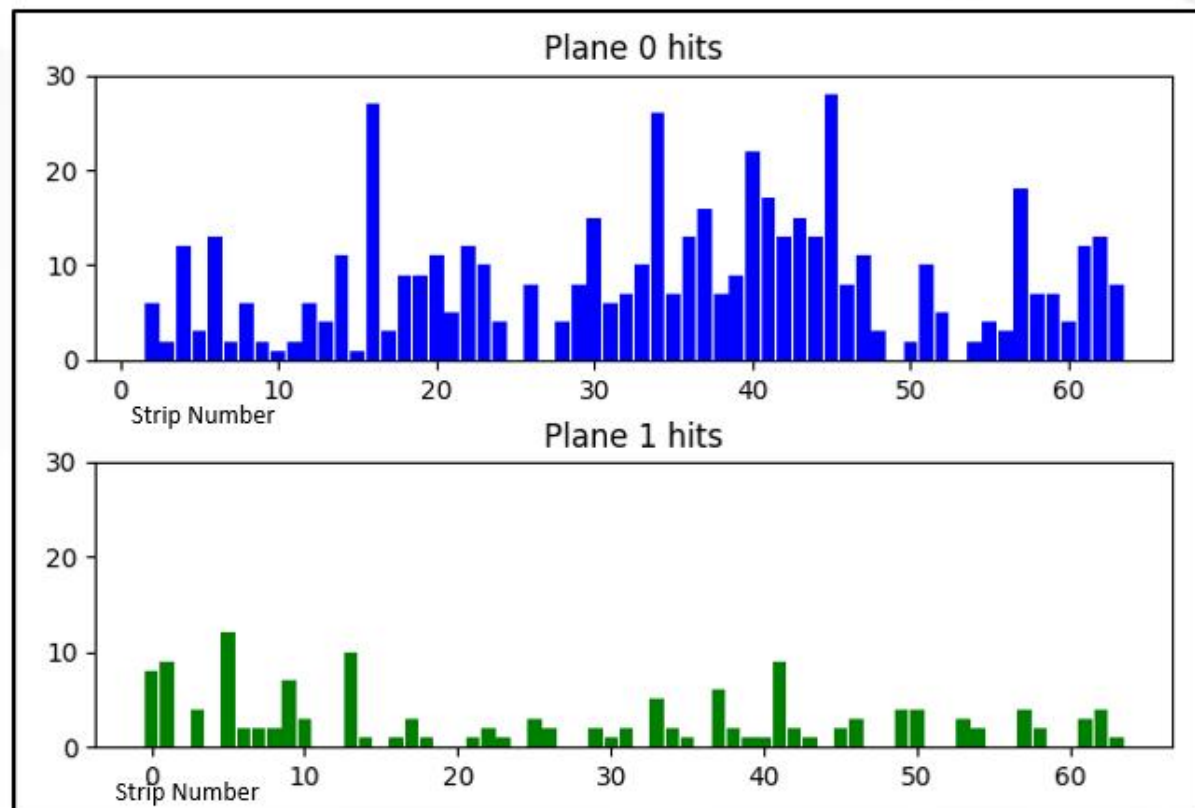


Closeup of interleaving

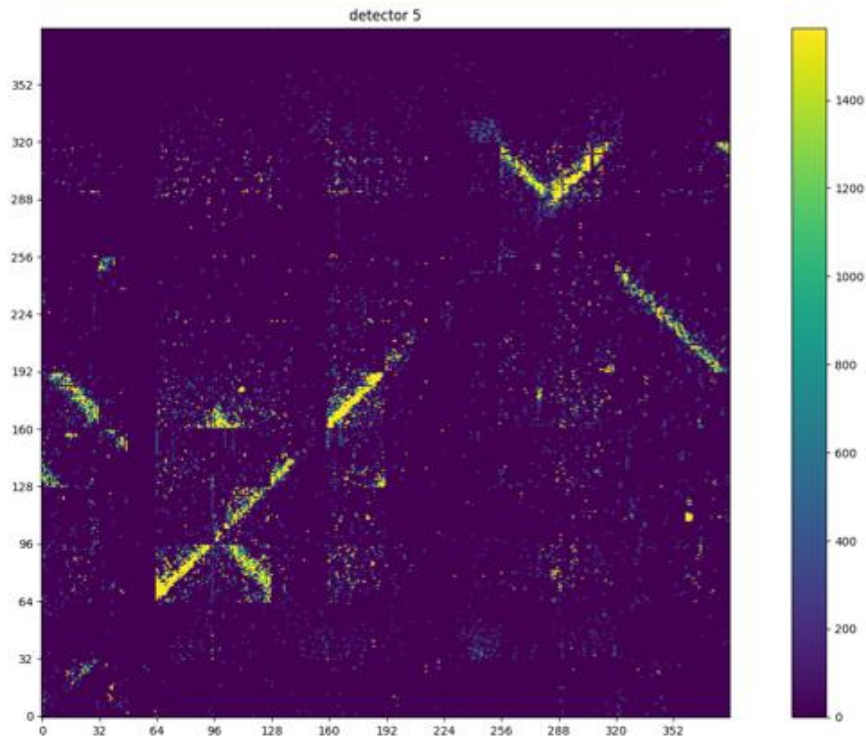


Order: 0, 127, 1, 126, 2, 125, ...

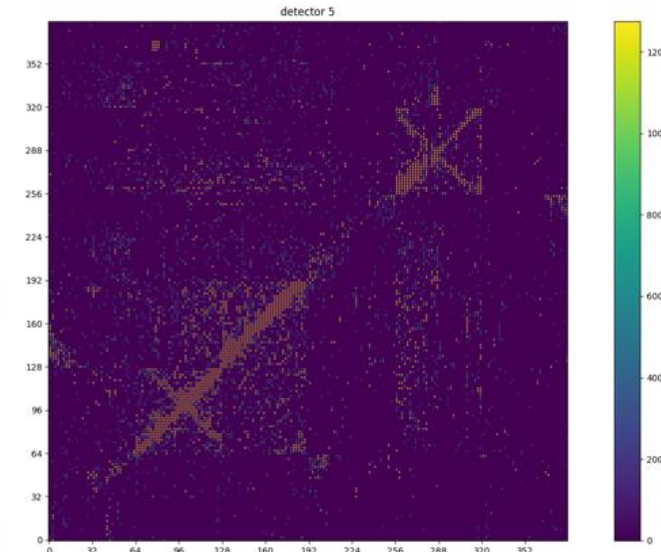
Approach- Generate mapping functions



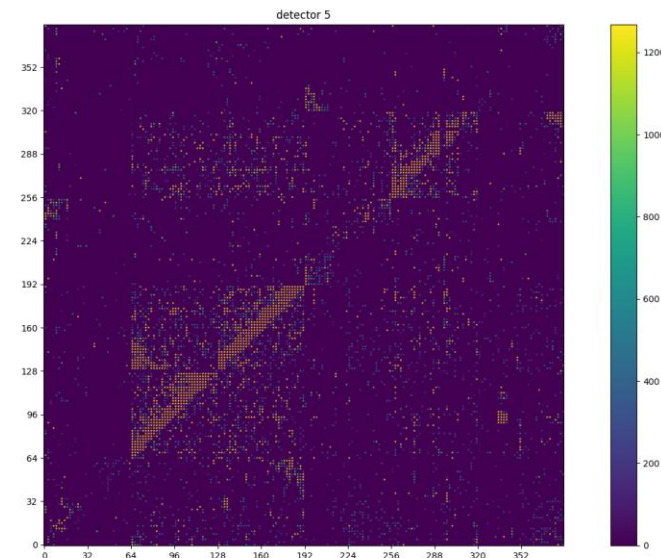
Approach- Mapping



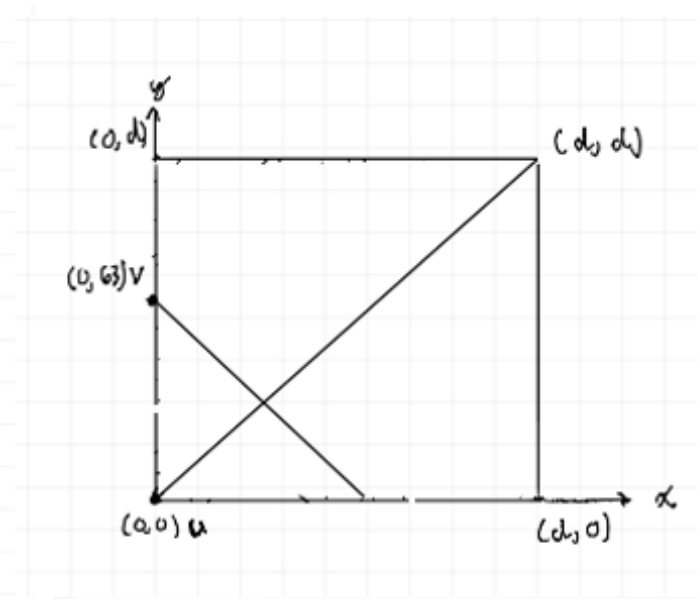
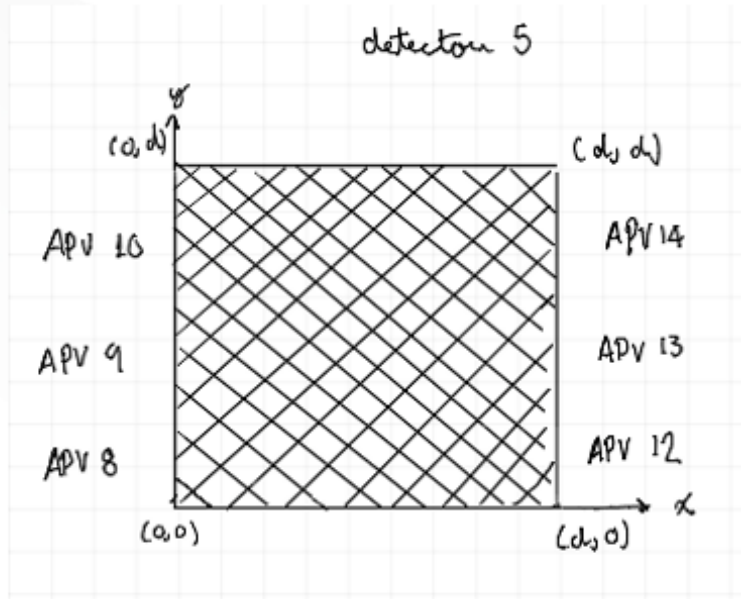
Before applying any mapping functions



After applying mapping functions



Approach- Resolution Plot



$$p_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad d_1 = u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$p_2 = \begin{bmatrix} 0 \\ g \end{bmatrix} \quad d_2 = v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$l_1 = p_1 + t d_1 \quad l_2 = p_2 + \lambda d_2$$

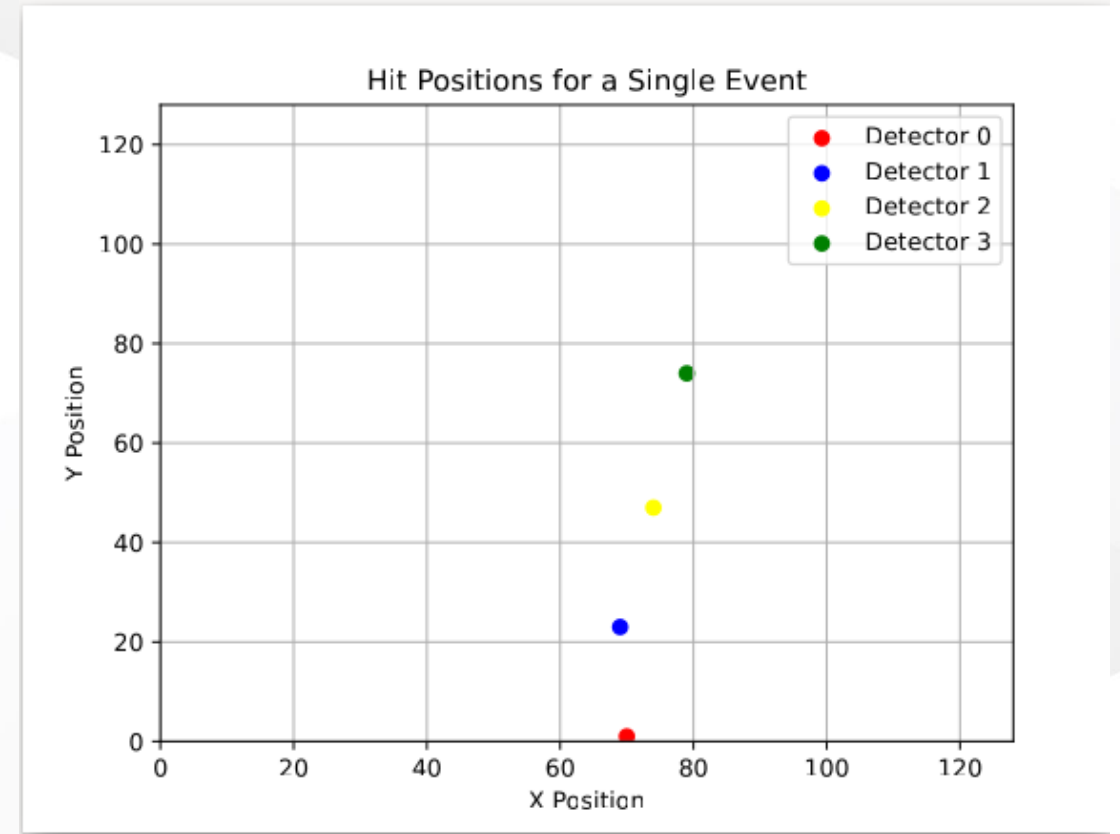
$$p_1 + t d_1 = p_2 + \lambda d_2$$

$$t d_1 - \lambda d_2 = p_2 - p_1$$

$$\begin{bmatrix} d_{1x} & -d_{2x} \\ d_{1y} & -d_{2y} \end{bmatrix} \begin{bmatrix} t \\ \lambda \end{bmatrix} = \begin{bmatrix} p_{2x} - p_{1x} \\ p_{2y} - p_{1y} \end{bmatrix}$$

Approach- Efficiency Plot

- Efficiency is used to evaluate the correctness of the cylindrical detector compared to the four other detectors in identifying the hit positions.
- We will compare the positions of hits from the same event across trackers and the detector.
- We will calculate and account for offset.



Novel Features

- Streamlined ability to take multiple beam data files
- Application of the correct pin to physical strip mapping despite nontrivial orientation
- Visualization of electron cluster location on the detector
- Thorough, detailed efficiency and resolution analysis

Algorithms and Tools

- Python: new codebase
- ROOT: physics analysis library for C++ and Python
- AMORE-SRS tools: framework for online monitoring of the particle data

Evaluation

- Speed- can each root file be processes in under two minutes?
- Accuracy- do are results for cylindrical data align with the planar tracker data?
- Reliability- does the system fail with corrupted data?
- User friendliness- can our stakeholders use and understand the software?

Progress Summary

Feature	Completion	To do
Perform reconstruction on planar data (from trackers)	100%	Nothing
Implement mathematical detector model to find intersecting strips	100%	Nothing
Generate pulse plots	100%	Nothing
Detector 5 mapping	70%	We have to find the correct mapping of 2 remaining APVs
Compare cylindrical detectors and trackers	15%	We have to invest methods for handling the offset
Create efficiency plots	0%	We haven't got the mapping to be able to create efficiency plots

Technical Challenges

- APVs from the cylindrical detector seem to be not in the order that we expected
- We have no way to access the APVs to see the orientation

Milestone 4

- Do beam reconstruction with different mappings so that we can find the correct one
- Compare different mappings against the correct tracker data
- Create efficiency plots for the different mappings

Milestone 5

- Do beam reconstruction on multiple root files with all detectors and trackers
- Create efficiency plots for multiple root files with all detectors and trackers.
- Automating the generation of efficiency plots against other relevant performance characteristics.
- Automating the generation of resolution plots to display the positions.

Milestone 6

- Write a paper about the work done.
- Do a demo of the pipeline and automatization features.
- Evaluate the results.
- Create documentation of the CLI

Milestone 4 Task Matrix

Task	Alejandro	Summer
Find correct cylindrical mapping	50%	50%
Create efficiency plots	50%	50%
Compare cylindrical detectors and trackers	50%	50%

Thank you for listening

Questions?